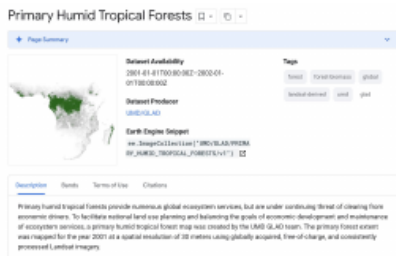
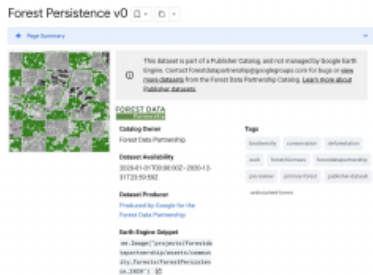


FOREST BASELINE

We integrate two forest baselines as the natural-forest reference against which change is measured: data from the University of Maryland (UMD) and the Forest Data Partnership. Both datasets are accessed through the Google Earth Engine (GEE) catalog.



1. Primary Humid Tropical Forest

Produced by UMD/GLAD

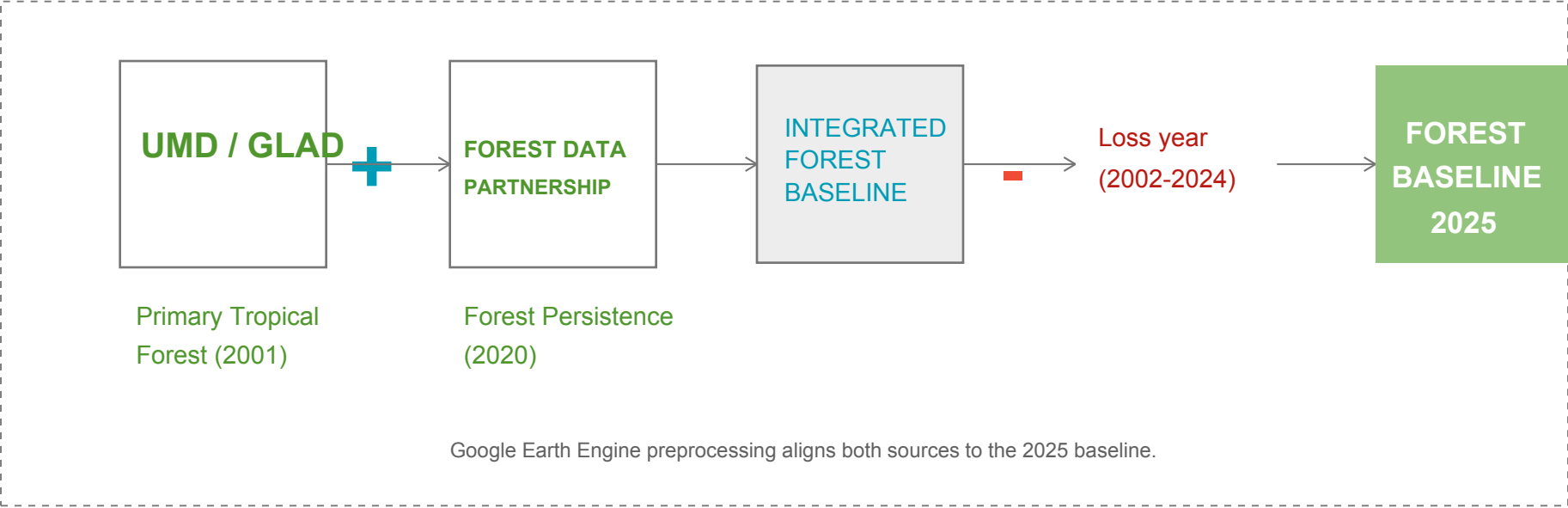
```
var dataset = ee.ImageCollection(  
'UMD/GLAD/PRIMARY_HUMID_TROPICAL_FORESTS/v1')  
.mosaic().selfMask();
```

2. Forest Persistence v0

Produced by Google for the Forest Data Partnership

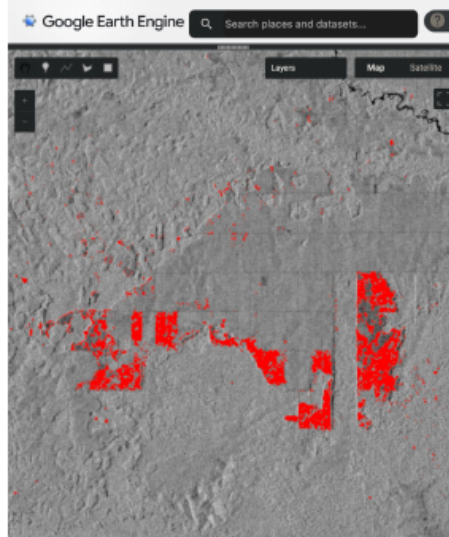
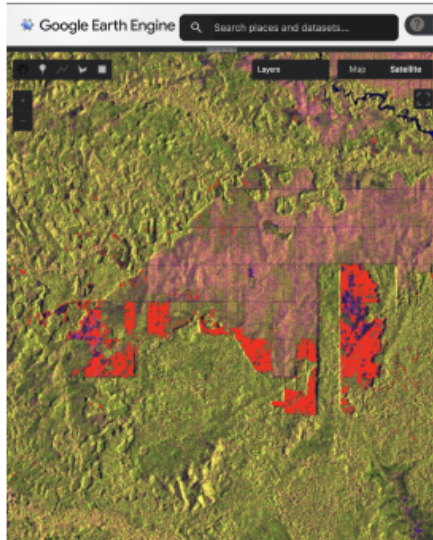
```
var dataset = ee.Image(  
'projects/forestdatapartnership/assets/community_forests/  
ForestPersistence_2020');
```

The two source baselines refer to different years. To detect forest change or loss alerts, each dataset is preprocessed to the required reference year. In this system, the alert-detection period starts in January 2026.



FOREST LOSS WARNINGS (ALERTS)

The system combines GLAD-L and RADD alerts for warning detection. RADD alerts are derived from Sentinel-1 radar imagery at 10-meter resolution, providing cloud-free remote-sensing data. GLAD alerts are derived from Landsat optical imagery at 30-meter resolution.



RADD (Radar for Detecting Deforestation)

is a near-real-time deforestation early-warning system that uses the European Space Agency's Sentinel-1 radar sensor to detect forest disturbance in primary tropical rainforests.

Developed by Wageningen University & Research (WUR), it is especially effective in tropical regions because Sentinel-1 radar penetrates clouds and records frequently, every 6-12 days. This makes it more consistent than optical-image systems.

Under ideal conditions, GLAD alerts can update about every eight days by combining Landsat 8 and Landsat 9 orbits. In cloud-prone tropical regions, however, the effective detection interval can be longer and depends on the availability of cloud-free imagery. Combining GLAD with cloud-independent RADD alerts strengthens early-warning detection for forest monitoring.

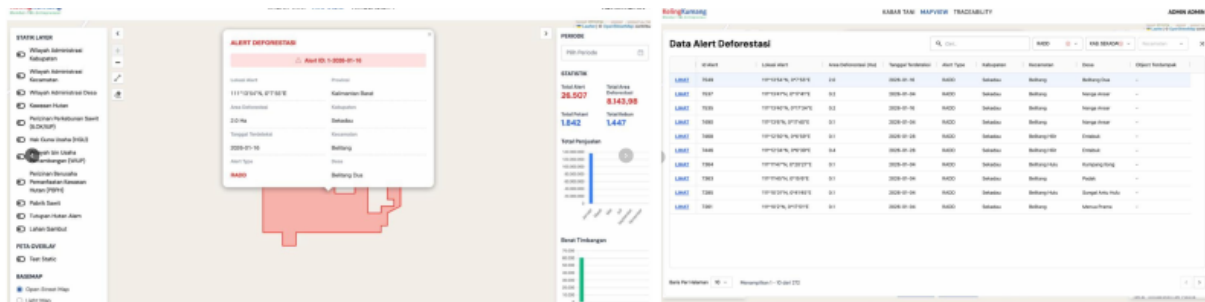
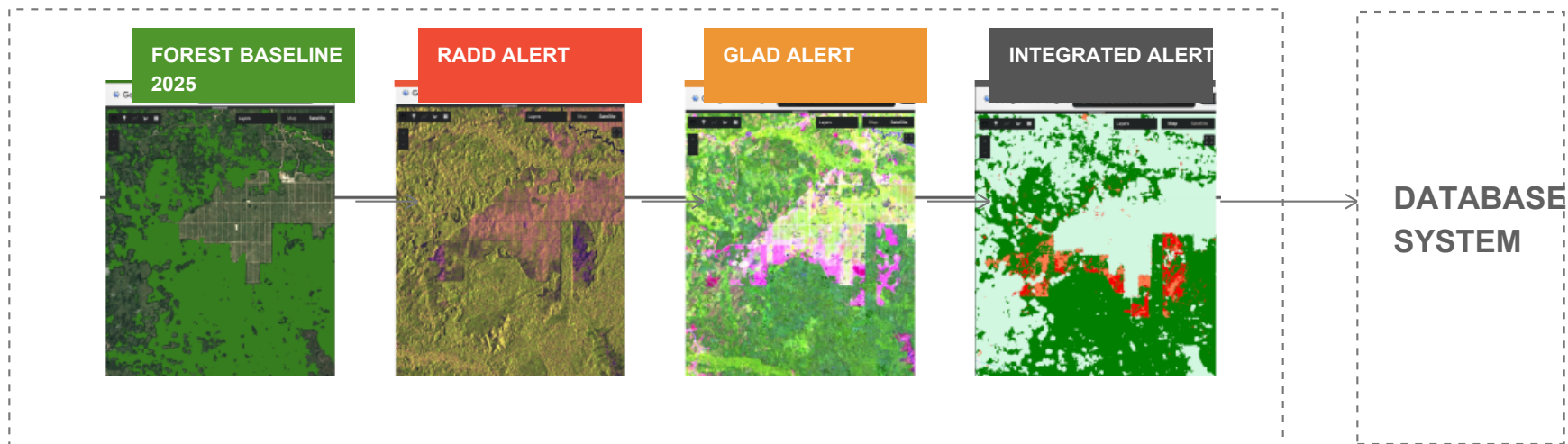


GLAD (Global Land Analysis and Discovery)

was developed by the University of Maryland using Landsat 8 and Landsat 9 imagery. It uses optical sensors to detect changes in tree cover from surface reflectance, with weekly updates when cloud-free imagery is available.

GLAD has a 30-meter spatial resolution; each pixel represents approximately 0.09 hectares. It is designed to detect forest-canopy disturbance caused by human activity or natural factors, including logging, agriculture, mining, and fire.

ALERT PROCESSING



ALERT INTEGRATION TRACEABILITY SYSTEM

Every detected alert is automatically sent to the system database and overlaid with thematic layers describing time, location, and affected area.